

Rendering Under Constraint

On the Selection of the Rendering Map in Emergent Boundary Geometry

Leslie Yarbrough

Independent Researcher

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Abstract

In prior work, temporal ordering was derived from memory topology and spatial geometry was proposed to emerge as a rendered image of constrained boundary structure. In such frameworks, a rendering map $F : T \times C \rightarrow \Sigma$ translates admissible boundary configurations into effective spatial geometry.

However, the introduction of a rendering map raises a critical structural question: what selects F ?

If the rendering process is unconstrained, the framework risks explaining too much while predicting too little. This note identifies minimal structural conditions that any physically meaningful rendering map must satisfy. These conditions restrict the admissible class of rendering maps without requiring a fully specified dynamical law.

The purpose of this note is therefore not to derive a unique rendering map, but to clarify the constraints under which a rendered-boundary framework can produce non-arbitrary spatial geometry.

1. Introduction

Recent work has explored the possibility that spacetime structure may be emergent rather than fundamental. In the author's earlier work [1], temporal ordering was derived from the topology of memory constraints rather than assumed as a primitive coordinate.

In subsequent work [2], spatial geometry was proposed to arise as a rendered image of a constrained boundary manifold. In that formulation, a boundary product space $T \times C$ is restricted by a compatibility manifold $M \subset T \times C$, and experienced spatial geometry Σ is generated through a rendering map

$$F : T \times C \rightarrow \Sigma$$

such that

$$\Sigma = F(M).$$

This formulation raises an immediate structural question. If F is arbitrary, the framework loses explanatory power. Any spatial geometry could be obtained by choosing a suitable rendering map.

The present note addresses this issue by identifying the minimal constraints that a physically meaningful rendering map must satisfy.

2. Structural Role of the Rendering Map

The rendering map F performs the transition from admissible boundary configurations to experienced spatial geometry.

Conceptually, the boundary manifold represents the space of possible configurations permitted by the constraint structure M . The rendering map then converts these configurations into geometric relationships within Σ .

Thus the rendering process must accomplish three tasks:

- translate boundary compatibility relations into spatial relations,
- preserve the structural constraints encoded in M ,
- produce a geometry that remains stable under small perturbations of the boundary configuration.

If these conditions are not met, the rendered geometry would either fail to reflect the underlying constraint structure or become physically unstable.

3. Minimal Constraints on the Rendering Map

We propose that any admissible rendering map must satisfy the following structural constraints.

3.1 Constraint Compatibility

The rendering map must preserve admissibility relations induced by the constraint manifold M . If two boundary configurations are incompatible under M , their rendered images should not appear as physically accessible spatial configurations.

This requirement ensures that the rendered geometry reflects the structure of the constraint manifold rather than overriding it.

3.2 Continuity and Stability

Small perturbations in boundary configuration should produce correspondingly small perturbations in rendered geometry.

Formally, this suggests that the rendering map should satisfy a continuity condition such as Lipschitz continuity,

$$\|F(x) - F(y)\| \leq k\|x - y\|.$$

Such a condition ensures that the rendering process does not introduce unphysical discontinuities in spatial geometry.

3.3 Minimal Distortion

The rendering map should preserve local relational structure as much as possible. In geometric terms, F should minimize distortion between neighboring boundary configurations and their spatial images.

While the precise metric notion of distortion remains to be specified, this condition expresses the intuitive requirement that the rendering process not arbitrarily scramble relational information.

3.4 Observer Equivalence

In earlier work, the coordinate C was interpreted as an observation interface through which boundary configurations are sampled.

Reparameterizations of this coordinate

$$\phi : C \rightarrow C$$

should not alter physical predictions. Therefore, the rendering map must be invariant under admissible reparameterizations of the interface.

4. Consequences for Emergent Geometry

These constraints significantly restrict the class of admissible rendering maps. Rather than acting as an arbitrary projection from boundary space to geometry, F must respect compatibility, continuity, and observer-equivalence conditions.

Within such a constrained class of maps, the resulting geometry is determined primarily by the structure of the constraint manifold M rather than by arbitrary choice of rendering rule.

This strengthens the interpretation of spatial geometry as an emergent property of boundary structure.

5. Conclusion

A rendered-boundary framework for emergent geometry requires that the rendering map F be constrained rather than arbitrary.

The present note identifies minimal structural conditions—constraint compatibility, continuity, minimal distortion, and observer equivalence—that any admissible rendering map must satisfy.

These conditions do not uniquely determine F , but they restrict the space of possible rendering rules sufficiently to preserve explanatory power.

Together with prior work on temporal ordering and boundary geometry, this analysis contributes to a broader program in which spacetime structure emerges from underlying constraint topology rather than existing as a fundamental background.

References

- [1] L. Yarbrough. *Time as Emergent from Memory Topology*. Preprint, 2025.
- [2] L. Yarbrough. *Space as Rendered Boundary*. Preprint, 2025.